

Contents

1	Measurable Functions	2
1.1	Elementray Properties of Measruable Functions	2

Measurable Functions

1.1 Elementary Properties of Measurable Functions

Definition 1.1.1

We endow $\overline{\mathbb{R}}$ with a topology called the order topology in the following manner. For each $a \in \mathbb{R}$ let

$$L_a = \overline{\mathbb{R}} \cap \{x : x < a\} = [-\infty, a) \quad \text{and} \quad R_a = \overline{\mathbb{R}} \cap \{x : x > a\} = (a, \infty].$$

The collection $\mathcal{S} = \{L_a : a \in \mathbb{R}\} \cup \{R_a : a \in \mathbb{R}\}$ is taken as a subbasis for this topology. A basis for the topology is given by

$$\mathcal{S} \cup \{R_a \cap L_b : a, b \in \mathbb{R}, a < b\}.$$

Observe that the topology on $\mathbb{R}$ induced by the order topology on $\overline{\mathbb{R}}$ is precisely the usual topology on $\mathbb{R}$.

Definition 1.1.2 ▶ Measurable Functions

- Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measure spaces. A mapping $f : X \rightarrow Y$ is called **measurable with respect to $\mathcal{M}$ and $\mathcal{N}$** if $f^{-1}(E) \in \mathcal{M}$ whenever $E \in \mathcal{N}$. If there is no danger of confusion, reference to $\mathcal{M}$ and $\mathcal{N}$ will be omitted, and we will simply use the term “measurable mapping.”
- If Y is a topological space, a restriction is placed on $\mathcal{N}$. In this case it is always assumed that $\mathcal{N}$ is the σ -algebra of Borel sets $\mathcal{B}$. Thus, in this situation, a mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable if

$$f^{-1}(E) \in \mathcal{M} \text{ whenever } E \in \mathcal{B}.$$

- When Y is taken as $\overline{\mathbb{R}}$ (endowed with the order topology) and $(X, \mathcal{M})$ is a topological space with $\mathcal{M}$ the collection of Borel sets. Then f is called a **Borel measurable function**.

- When $X = \mathbb{R}^n$, $\mathcal{M}$ is the class of Lebesgue measurable sets and $Y = \overline{\mathbb{R}}$. Here, the measurability of f required that $f^{-1}(E)$ be Lebesgue measurable whenever $E \subset \overline{\mathbb{R}}$ is Borel, in which case f is called a **Lebesgue measurable function**. The definitions imply that E is a measurable set if and only if χ_E is a measurable function.

Theorem 1.1.3

If the mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable, where $\mathcal{B}$ is the σ -algebra of Borel sets. Define

$$\Sigma = \{E : E \subset Y \text{ and } f^{-1}(E) \in \mathcal{M}\}.$$

Then Σ is a σ -algebra.

Proof. Easy to verify. □

Corollary 1.1.4

A continuous mapping is a Borel measurable function.

Proof. If f is continuous, Σ contains all open sets. Since Σ is a σ -algebra, Σ contains all Borel sets. □

Definition 1.1.5

If $f : X \rightarrow \mathbb{R}$, we call the sets

$$\{f > a\} := \{x \in X : f(x) > a\}$$

the **superlevel sets of f** .

Theorem 1.1.6

Let $f : X \rightarrow \overline{\mathbb{R}}$, where $(X, \mathcal{M})$ is a measure space. The following conditions are equivalent:

- (i) f is measurable.
- (ii) $\{f > a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iii) $\{f \geq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
- (iv) $\{f < a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

(v) $\{f \leq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

Theorem 1.1.7

A function $f : X \rightarrow \overline{\mathbb{R}}$ is measurable if and only if

- (i) $f^{-1}\{-\infty\} \in \mathcal{M}$ and $f^{-1}\{\infty\} \in \mathcal{M}$ and
- (ii) $f^{-1}(a, b) \in \mathcal{M}$ for all open intervals $(a, b) \subset \mathbb{R}$.

Lemma 1.1.8

If f and g are measurable functions, then the following sets are measurable:

- (i) $X \cap \{x : f(x) > g(x)\}$,
- (ii) $X \cap \{x : f(x) \geq g(x)\}$,
- (iii) $X \cap \{x : f(x) = g(x)\}$.

Proof. If $f(x) > g(x)$, then there is a rational number r such that $f(x) > r > g(x)$. Therefore, it follows that

$$\{f > g\} = \bigcup_{r \in \mathbb{Q}} (\{f > r\} \cap \{g < r\}),$$

which is a measurable set, and easily (i) holds. (ii) is just the complements of (i) with f and g interchanged. (iii) is just the intersection two of two measurable sets of type (ii), and so it too is measurable. $\square$

Definition 1.1.9

If f and g are real-valued measurable functions, then $f + g$ is undefined at points where it would be of the form $\infty - \infty$. This difficulty is overcome if we define

$$(f + g)(x) := \begin{cases} f(x) + g(x), & x \in X - B, \\ \alpha, & x \in B, \end{cases}$$

where $\alpha \in \overline{\mathbb{R}}$ is chosen arbitrarily and where

$$B := (f^{-1}\{\infty\} \cap g^{-1}\{-\infty\}) \cup (f^{-1}\{-\infty\} \cap g^{-1}\{\infty\}).$$

Theorem 1.1.10

If $f, g : X \rightarrow \overline{\mathbb{R}}$ are measurable functions, then $f + g$ and fg are measurable.

Proof. We will treat the case that f and g have values in $\mathbb{R}$. The proof is similar in the general case and is left as an exercise (see Exercise 2, Section 5.1).

To prove that the sum is measurable, define $F : X \rightarrow \mathbb{R} \times \mathbb{R}$ by

$$F(x) = (f(x), g(x))$$

and $G : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ by

$$G(x, y) = x + y.$$

Then $G \circ F(x) = f(x) + g(x)$, so it suffices to show that $G \circ F$ is measurable. Referring to Theorem 1.1.7, we need only show that $(G \circ F)^{-1}(J) \in \mathcal{M}$ whenever $J \subset \mathbb{R}$ is an open interval. Now $U := G^{-1}(J)$ is an open set in $\mathbb{R}^2$, since G is continuous. Furthermore, U is the union of a countable family, $\mathcal{F}$, of 2-dimensional intervals I of the form $I = I_1 \times I_2$, where I_1 and I_2 are open intervals in $\mathbb{R}$. Since

$$F^{-1}(I) = f^{-1}(I_1) \cap g^{-1}(I_2),$$

we have

$$F^{-1}(U) = F^{-1}\left(\bigcup_{I \in \mathcal{F}} I\right) = \bigcup_{I \in \mathcal{F}} F^{-1}(I),$$

which is a measurable set. Thus $G \circ F$ is measurable, since

$$(G \circ F)^{-1}(J) = F^{-1}(U).$$

The product is measurable by essentially the same proof. □

Definition 1.1.11 ▶ Cantor Function

Recall that the Cantor set C can be expressed as

$$C = \bigcap_{j=1}^{\infty} C_j$$

Where C_j is the union of the 2^j closed intervals that remain after the j th step of the construction. The set

$$D_j = [0, 1] - C_j$$

consists of the 2^{j-1} open intervals that are deleted at the j th step. Let these intervals be denoted by $I_{j,k}$, $k = 1, 2, \dots, 2^{j-1}$, and order them in the obvious way from left to right. Now define a continuous function f_j on $[0, 1]$ by

$$\begin{aligned} f_j(0) &= 0, \\ f_j(1) &= 1, \\ f_j(x) &= \frac{k}{2^j}, \quad x \in I_{j,k} \end{aligned}$$

and define f_j linearly on each interval of C_j . The function f_j is continuous and nondecreasing, and it satisfies

$$|f_j(x) - f_{j+1}(x)| < \frac{1}{2^j}, \quad x \in [0, 1]$$

Since

$$|f_j - f_{j+m}| < \sum_{i=j}^{j+m-1} \frac{1}{2^i} < \frac{1}{2^{j-1}}$$

it follows that the sequence $\{f_j\}$ converges uniformly to a continuous function f , called **Cantor-Lebesgue function**.

Note.

f_j 在 $I_{j,k}$ 上的设定是端点的平均值.

Corollary 1.1.12

The Cantor-Lebesgue function f and its associate $h(x) := f(x) + x$ described above have the following properties:

- (i) $f(C) = [0, 1]$; that is, f maps a set of measure 0 onto a set of positive measure.
- (ii) h maps a Lebesgue measurable set onto a nonmeasurable set.
- (iii) The composition of Lebesgue measurable functions need not be Lebesgue measurable.

(iv) Lebesgue measurable functions do not preserve Borel sets.

Proof. 1. $f : [0, 1] \rightarrow [0, 1]$ is onto. It is easy to see that $f(C) = [0, 1]$, because $f(C)$ is compact and $f([0, 1] - C)$ is countable.

2.

Proof sketch.

把 C 看成是集合 $[0, 1]$ 测度为零的“骨”. $[0, 1] \setminus C$ 是 $[0, 1]$ 的测度为 1 的“肉”. f 的构造是对“骨”的展开: $f(C) = [0, 1]$; 同时对“肉”的压缩: 令 f 在 $I_{j,k}$ 上取常值. 通过令 $h : [0, 1] \rightarrow [0, 2]$, $h(x) = f(x) + x$, 我们变换映射的尺度, 不再将“肉”压缩, 而是忠实地搬运, 将每个 $I_{j,k}$ 映到大小相同的一个区间. 从而间接的看到这两个操作对“骨”的影响区别不大 (测度都是 1).

Note.

$h(C)$ 和 $f(C)$ 测度相同在直觉上也是存在依据的.

Cantor 函数 f 是一列分段线性函数 f_n 的一致极限. 在构成 Cantor 集第 n 步近似 C_n 的那些区间上, f_n 的斜率为 $(3/2)^n$, 这是一个趋向于无穷大的值.

函数 $h(x) = f(x) + x$ 可以看作是 $h_n(x) = f_n(x) + x$ 的极限. 在同样的区间上, h_n 的斜率变为 $(3/2)^n + 1$.

为一个趋于无穷的量加上 1, 并不会显著改变它的数量级. 两个斜率的比值为 $\frac{(3/2)^{n+1}}{(3/2)^n} = 1 + (2/3)^n \rightarrow 1$.

这便解释了我们的直觉: “+x” 这个操作虽然修复了 f 对 C 补集测度的“压缩”, 但对于 f 自身在 C 上的“测度生成”效应来说, 其 * 相对 * 影响是微不足道的. 因此, f 和 h 都将测度为零的集合 C “拉伸”成了一个测度为 1 的集合.

h is strictly increasing. Thus, h is a homeomorphism from $[0, 1]$ onto $[0, 2]$. Then h carries the open set $[0, 1] \setminus C$ to an open set. And we have

$$\begin{aligned} m(h([0, 1] \setminus C)) &= m\left(h\left(\bigcup_{j,k} I_{j,k}\right)\right) \\ &= \sum_{j,k} m(h(I_{j,k})) \\ &= \sum_{j,k} m(I_{j,k}) \\ &= 1 \end{aligned}$$

Therefore, h maps the Cantor set to a set P with measure 1. From Exercise ??, there exists a non-measurable set $N \subseteq P$. Set $A = h^{-1}(N)$. Since $h^{-1}(N) \subseteq h^{-1}(P) = C$, A is a subset of measure-zero set C . Thus A is Lebesgue-measurable. But $h(A) = N$ is not measurable.

3. Note that h^{-1} is measurable, since it is continuous. Let $F := h^{-1}$. Observe that χ_A is measurable, but

$$\chi_A \circ F = \chi_A \circ h^{-1} = \chi_N$$

is not Lebesgue-measurable.

4. Observe that A is not a Borel set, for if it were, then $F^{-1}(A)$ would be a Borel set. But $F^{-1}(A) = h(A) = N$ and N is not a Borel set. Now $g := \chi_A$ is a Lebesgue measurable function such that $g^{-1}(\{1\}) = A$, where $\{1\}$ is a Borel set but A is not.

□

Theorem 1.1.13

Suppose $f : X \rightarrow \overline{\mathbb{R}}$ is measurable and $g : \overline{\mathbb{R}} \rightarrow \overline{\mathbb{R}}$ is Borel measurable. Then $g \circ f$ is measurable. In particular, if $X = \mathbb{R}^n$ and f is Lebesgue measurable, then $g \circ f$ is Lebesgue measurable.

Corollary 1.1.14

Let $f : X \rightarrow \overline{\mathbb{R}}$ be a measurable function.

- (i) Let $\varphi(x) = |f(x)|^p$, $0 < p < \infty$, and let φ assume arbitrary extended values on the sets $f^{-1}(\infty)$ and $f^{-1}(-\infty)$. Then φ is measurable.
- (ii) Let $\varphi(x) = \frac{1}{f(x)}$, and let φ assume arbitrary extended values on the sets $f^{-1}(0)$, $f^{-1}(\infty)$ and $f^{-1}(-\infty)$. Then φ is measurable.

In particular, if $X = \mathbb{R}^n$ and f is Lebesgue measurable, then φ is Lebesgue measurable in (i) and (ii).

Proof. 1. Define $g(t) = |t|^p$ for $t \in \mathbb{R}$, and assign arbitrary values to $g(\infty)$ and $g(-\infty)$. Now apply the previous theorem.

2. Proceed in a similar way by defining $g(t) = \frac{1}{t}$ when $t \neq 0, \infty, -\infty$ and assigning arbitrary values to $g(0)$, $g(\infty)$, and $g(-\infty)$.

□

Theorem 1.1.15

Let $(X, \mathcal{M}, \mu)$ be a complete measure space and let f, g be extended real-valued functions defined on X . If f is measurable and $f = g$ almost everywhere, then g is measurable.

Remark.

In a complete measure space, we can consider a function f that is just defined on almost everywhere on X , by extending f into $\bar{f}$ with arbitrary value taken on where f was not defined. .

Proof. Let $N = \{x : f(x) \neq g(x)\}$. Then $\mu(N) = 0$, and thus N as well as all subsets of N are measurable. For $a \in \mathbb{R}$, we have

$$\begin{aligned} \{g > a\} &= (\{g > a\} \cap N) \cup (\{g > a\} \cap N^c) \\ &= (\{f > a\} \cap N) \cup (\{f > a\} \cap N^c) \in \mathcal{M}. \end{aligned}$$

□

Theorem 1.1.16

Suppose φ is an outer measure on a space X . Then an extended real-valued function f on X is φ -measurable if and only if

$$\varphi(A) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f \geq b\})$$

whenever $A \subset X$ and $a < b$ are real numbers.

Proof. The “only if” is immediate from

$$\varphi(A) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f > a\}) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f \geq b\})$$

To show the converse, we need to prove that for any $r \in \mathbb{R}$, the set

$$E := \{f \leq r\}$$

satisfies

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E)$$

for each $A \subseteq X$. We define

$$B_j := A \cap \left\{ r + \frac{1}{j+1} \leq f \leq r + \frac{1}{j} \right\}$$

We shall show

$$\infty > \varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j}\right) = \sum_{j=1}^{\infty} \varphi(B_{2j})$$

by induction. We fix k and assume

$$\varphi\left(\bigcup_{j=1}^k B_{2j}\right) = \sum_{j=1}^k \varphi(B_{2j})$$

We set

$$A_k := \bigcup_{j=1}^k B_{2j}$$

Then

$$(A_k \cup B_{2(k+1)}) \cap \left\{f \leq r + \frac{1}{2(k+1)}\right\} = B_{2(k+1)}$$

$$(A_k \cup B_{2(k+1)}) \cap \left\{f \geq r + \frac{1}{2k+1}\right\} = A_k$$

We have

$$\varphi\left(\bigcup_{j=1}^{k+1} B_{2j}\right) = \varphi(A_k \cup B_{2(k+1)}) \geq \varphi(B_{2(k+1)}) + \varphi(A_k) = \sum_{j=1}^{k+1} \varphi(B_{2j})$$

Thus we inductively show that

$$\varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j}\right) = \sum_{j=1}^{\infty} \varphi(B_{2j})$$

Similarly, one can show that

$$\varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j-1}\right) = \sum_{j=1}^{\infty} \varphi(B_{2j-1})$$

Then the series

$$\sum_{j=1}^{\infty} \varphi(B_j) \leq 2\varphi(A) < \infty$$

converges. The tail of the series can be arbitrarily small. For each $\varepsilon > 0$, there exists $m \in \mathbb{N}$ such that

$$\varepsilon > \sum_{j=m}^{\infty} \varphi(B_j) \geq \varphi\left(\bigcup_{j=m}^{\infty} B_j\right)$$

Where

$$\bigcup_{j=m}^{\infty} B_j = A \cap \left\{r < f \leq r + \frac{1}{m}\right\}$$

We define a measure ψ

$$\psi(F) := \varphi(A \cap F), \quad \forall F \subseteq X$$

Then

$$\varepsilon > \psi\left(\left\{r < f \leq r + \frac{1}{m}\right\}\right)$$

Thus

$$\begin{aligned} \varphi(A \cap E) + \varphi(A - E) &= \psi(E) + \psi(E^c) \\ &= \psi(\{f \leq r\}) + \psi(\{f > r\}) \\ &\leq \psi(\{f \leq r\}) + \psi\left(\left\{f > r + \frac{1}{m}\right\}\right) + \psi\left(\left\{r < f \leq r + \frac{1}{m}\right\}\right) \\ &\leq \psi(\{f \leq r\}) + \psi\left(\left\{f \geq r + \frac{1}{m}\right\}\right) + \varepsilon \\ &= \varphi(A \cap \{f \leq r\}) + \varphi\left(A \cap \left\{f \geq r + \frac{1}{m}\right\}\right) + \varepsilon \\ &\leq \varphi(A) + \varepsilon \end{aligned}$$

Since ε is arbitrary, we have

$$\varphi(A \cap E) + \varphi(A - E) \leq \varphi(A)$$

Thus E is measurable. □